# PINO: Person-Interaction Noise Optimization for Long-Duration and Customizable Motion Generation of Arbitrary-Sized Groups

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## Abstract

*Generating text-driven human interactions beyond two people remains difficult because interaction complexity grows combinatorially with group size, while existing models are typically trained for a fixed number of actors and often produce collisions, shared-prompt ambiguity, or limited spatial control. We present PINO, a training-free framework that composes arbitrary-sized group interactions from a pretrained two-person diffusion model by optimizing the initial diffusion noise at inference time. PINO adds one new character at a time around a shared pivot person, reusing the pretrained model through masked conditioning so previously generated motion is preserved while pair-specific prompts define the role of the newly added participant. The optimization objective combines collision-aware and spatio-temporal penalties, enabling overlap avoidance, root-position control, region constraints, and orientation control without retraining the backbone model. PINO also extends interactions to longer durations through inpainting-based continuation with optimized unobserved segments. When instantiated on InterGen, PINO reduces overlap from 0.119 to 0.000 and penetration volume from 3112.72 to 275.65 in two-person generation while preserving semantic quality. In multi-person generation, it reduces overlap from 0.766 to 0.000 and penetration volume from 25215.78 to 785.01, and it continues to scale favorably up to five people. For motion extension, PINO improves both semantic fidelity and physical plausibility, reducing FID from 20.390 to 18.930 and overlap from 0.089 to 0.000. These results show that inference-time noise optimization is a practical route to controllable and physically coherent group interaction synthesis without additional training.*

## 1. Introduction

Text-driven human motion generation has advanced rapidly with diffusion models for single-person motion and two-person interaction synthesis [2, 6, 11, 14]. However, generating realistic interactions for arbitrary-sized groups remains substantially harder than the two-person case. The main difficulty is not only modeling additional bodies, but preserving coherent semantics while preventing trivial failure modes such as body overlap, interpenetration, and role ambiguity across multiple participants.

Existing approaches address this challenge from two main directions. One line trains dedicated multi-person generators that directly model several people at once [1, 9]. This offers expressive generation, but the architecture and training data are tied to the group distributions seen during training. Another line keeps pretrained priors fixed and applies inference-time control or optimization [5, 7, 12]. These methods improve flexibility, but they do not directly exploit the structure of human group interactions as a composition of localized pairwise relations.

Our starting observation is that many group interactions can be organized around a shared pivot person. For example, a central character may greet one person, guide another, and coordinate with several surrounding partners. This suggests a simple decomposition: instead of asking one model invocation to explain the entire group, we can sequentially compose pairwise interactions between the pivot and each newly added person. A pretrained two-person interaction diffusion model already contains a strong prior over pose realism, timing, and semantic interaction. The missing component is a mechanism that preserves this prior while enforcing group-level physical consistency and user control.

Based on this idea, we propose **PINO** (**P**erson-**I**nteraction **N**oise **O**ptimization), a training-free framework for long-duration and customizable group interaction generation. PINO uses a pretrained two-person diffusion model as a frozen backbone, then optimizes the initial diffusion noise for each pairwise generation step. Masked conditioning keeps the pivot motion fixed, while the newly added character is synthesized from a pair-specific text prompt. The optimization objective is defined directly in motion space and includes penalties for overlap avoidance, start/end root positions, movement regions, and orientation. This retains the semantic prior of the pretrained model while giving the user precise inference-time control.

PINO also supports long-duration interaction generation. Rather than retraining a dedicated long-horizon model, we extend a motion sequence by treating known frames as fixed conditions and optimizing the unobserved segment

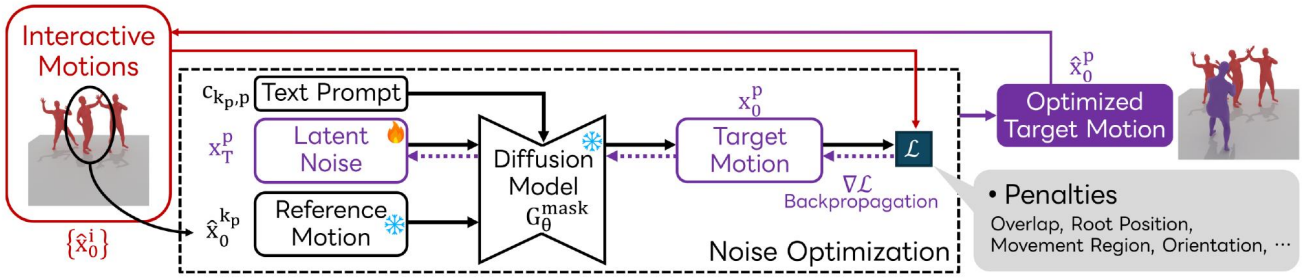


Figure 1. **Person-Interaction Noise Optimization (PINO).** Starting from a pretrained two-person diffusion model, PINO generates an initial pair and then sequentially adds new participants around a pivot person through masked conditioning. The initial noise for the new pair is optimized with physics-aware and spatio-temporal penalties, enabling collision avoidance and controllable group composition without retraining.

through inpainting. This preserves continuity while keeping the same inference-time control mechanism used for group composition.

Experiments on InterGen-based generation show a consistent pattern: PINO yields the largest gains in physical plausibility and scalability, while preserving or modestly improving semantic metrics. In two-person generation, PINO eliminates overlap and sharply reduces penetration volume without degrading FID. In multi-person generation, the gap becomes larger, especially as the number of people increases. In motion extension, PINO improves both semantic fidelity and non-semantic realism.

Our contributions are threefold. First, we introduce a training-free framework that composes arbitrary-sized group interactions from a pretrained two-person diffusion model through sequential pivot-based generation. Second, we formulate inference-time person-interaction noise optimization with collision-aware and spatio-temporal penalties, enabling customizable control without retraining. Third, we show that the same mechanism extends to long-duration interaction generation through inpainting-based continuation.

## 2. Related Work

### 2.1. Diffusion Models for Human Motion

Diffusion models have become a standard backbone for text-to-motion generation because they produce diverse and realistic trajectories while supporting flexible conditioning [3, 10, 11, 14]. Subsequent work improved fidelity and controllability through masked modeling, latent-space generation, or stronger priors [2, 8]. These methods focus primarily on single-person motion and do not directly address multi-person physical constraints. PINO instead treats a pretrained motion diffusion model as a reusable prior and moves the control problem to inference time.

### 2.2. Interactive and Multi-Person Motion Generation

Two-person interaction generation has recently become feasible with specialized architectures and datasets, especially through InterGen [6]. More recent interactive generators further improve fidelity and reaction modeling, but they still target a fixed number of participants during training. For larger groups, FreeMotion [1] and Shan et al. [9] move toward open-domain multi-person generation, yet they model multiple people jointly and therefore inherit the difficulty of scaling network design and training data to larger groups. PINO takes a different route: it reuses a pretrained two-person model and composes larger groups sequentially, which makes the number of generated people an inference-time choice rather than a training-time design decision.

### 2.3. Inference-Time Control and Optimization

Inference-time control has emerged as an attractive alternative to retraining motion generators. Guided Motion Diffusion [4] and OmniControl [13] incorporate spatial constraints for single-person motion generation. Programmable Motion Generation [7] and Diffusion Noise Optimization [5] show that optimizing latent variables or diffusion noise can satisfy diverse motion-space constraints without task-specific retraining. In the multi-person setting, InterControl [12] uses joint-level interaction control for zero-shot synthesis. PINO is closest in spirit to these training-free methods, but differs in two ways: it explicitly decomposes group generation into pivot-centered pairwise interactions, and it uses the optimization to preserve physical consistency among already generated participants while allowing prompt switching across pairs.

## 3. Method

### 3.1. Problem Setup

We assume access to a pretrained text-conditioned two-person diffusion model $G$ such as InterGen [6]. Given a text

| Method | #Person in Network Design | #Person in Training Data | Order of Multi-Person Generation | Method |
|---|---|---|---|---|
| InterControl [12] | One | One | Sequential | ControlNet |
| FreeMotion [1] | One | Two | Sequential | ControlNet |
| Shan et al. [9] | Multiple | Multiple | Concurrent | Supervised Training |
| ProgMoGen [7] | One | Training-free | N/A (Single-Person) | Noise Optimization |
| Ours | Two (or multiple) | Training-free | Sequential | Noise Optimization |

Table 1. Comparison of recent multi-person interaction generation methods, including single-person method ProgMoGen [7].

prompt $p$, the model maps an initial Gaussian noise tensor $z$ to a motion sequence $\hat{X} = G(z, p)$ containing two interacting people. Each person is represented by a sequence of motion features; in the InterGen instantiation, a single person uses 262 features per frame, and optimization is performed on 210-frame windows.

Our goal is to generate a group of $N$ interacting people, potentially with pair-specific prompts and motion constraints, without changing the pretrained weights of $G$. We also want the same mechanism to support motion extension beyond the native window length of the pretrained model.

### 3.2. Pivot-Based Pairwise Composition

The key design choice in PINO is to organize group generation around a *pivot* person shared across pairwise generations. We first sample a two-person interaction from prompt $p_{1,2}$ to obtain the pivot motion $X^{(1)}$ and the second participant motion $X^{(2)}$. To add a new participant $k$, we invoke the same two-person diffusion model again with a new prompt $p_{1,k}$, but now one input slot is fixed to the pivot motion:

$$\hat{X}^{(1,k)} = G(z_k, p_{1,k}; X^{(1)}, M), \tag{1}$$

where $M$ is a binary mask that preserves the pivot features and leaves the new participant unconstrained.

This masked conditioning is implemented through the inpainting interface of the frozen backbone. Concretely, the known coordinates of the pivot motion are copied into the denoiser output at every reverse step, so optimization only modifies the unconstrained participant. The result is a new motion $X^{(k)}$ that interacts with the pivot while remaining compatible with previously generated people. Repeating this process sequentially yields groups of arbitrary size.

This decomposition has two benefits. First, it avoids asking a single prompt to explain the roles of all participants at once. Each pair can have its own prompt, which allows prompt switching and clearer role assignment. Second, it leverages a pretrained interaction prior that already models close-range coordination, while group-level consistency is handled by optimization. The pivot assumption is also computationally useful. Keeping one actor fixed across generations propagates a stable temporal reference into every new pair, which reduces the number of unconstrained bodies that the optimizer must coordinate simultaneously. In practice, this turns a hard many-body search problem into a sequence of smaller constrained updates while still allowing non-pivot participants to interact indirectly through the shared spatial layout.

### 3.3. Person-Interaction Noise Optimization

For each generation step, PINO optimizes the initial diffusion noise rather than retraining or fine-tuning the backbone. Let $z_k^0$ be an initial random noise sample for the pair consisting of the pivot and a new target person. PINO solves

$$z_k^\star = \arg\min_{z_k} \ \lambda_{\text{ov}}\mathcal{L}_{\text{overlap}} + \lambda_{\text{root}}\mathcal{L}_{\text{root}} + \lambda_{\text{reg}}\mathcal{L}_{\text{region}} + \lambda_{\text{ori}}\mathcal{L}_{\text{orientation}}, \tag{2}$$

where all losses are computed on the decoded motion.

**Overlap loss.** The default objective encourages the newly generated person to maintain a valid distance from the pivot and from all previously generated non-pivot participants. In the released implementation, this is expressed through root-distance penalties on the pelvis trajectory, with a minimum distance threshold for collision avoidance and an optional maximum threshold to prevent degenerate separations. When generating the $k$-th person, the loss also pushes that person away from already generated people other than the pivot.

**Root-position loss.** To support temporal anchors, PINO can constrain the start and end root positions of one or both participants. This is useful when the user needs a character to enter or exit at a prescribed location, or when the motion must align with external timing constraints.

**Region loss.** PINO can restrict a participant to remain inside or outside a spatial region. The implementation supports circular or rectangular regions on the ground plane, enabling scene-aware placement or motion zoning without retraining a scene-conditioned model.

**Orientation loss.** An orientation penalty aligns the participant's facing direction with a target direction. This is especially useful for prompts whose semantics depend on mutual attention, such as facing a pivot before approaching or gesturing.

The optimization is fully differentiable because gradients propagate from the motion-space objective through the denoising process back to the initial noise. In practice, we use Adam for up to 100 iterations with early stopping, following the optimization loop implemented in the released code.

**Algorithm 1** PINO inference for group composition

Generate $(X^{(1)}, X^{(2)}) \leftarrow G(z_2, p_{1,2})$
**for** $k = 3$ to $N$ **do**
  Sample $z_k$ and fix pivot motion $X^{(1)}$ with mask $M$
  Optimize $z_k$ using Eq. (2)
  Decode $X^{(k)} \leftarrow G(z_k^\star, p_{1,k}; X^{(1)}, M)$
**end for**
**return** $\{X^{(i)}\}_{i=1}^{N}$

### 3.4. Prompt Switching and Customizable Roles

Because each pairwise generation step uses its own text prompt, PINO naturally supports prompt switching across participants. A single group motion can therefore mix semantically different relations around the same pivot, such as handshaking with one person while addressing or guiding another. This is difficult for shared-prompt approaches, which often blur individual roles when several people must simultaneously satisfy one global textual description.

### 3.5. Long-Duration Motion Extension

The same formulation can extend interactions beyond the native duration of the pretrained model. We keep a subset of frames fixed as observed motion, mask them through the inpainting interface, and optimize the noise for the remaining interval. This reuses the pretrained prior for local motion realism while allowing the unobserved portion to adapt to collision-aware and control-oriented objectives. In practice, the known segment acts as a temporal anchor at the transition boundary, producing longer interactions without training a separate long-horizon model.

## 4. Experiments

### 4.1. Experimental Setup

We instantiate PINO on InterGen [6] and denote the resulting system as *PINO-InterGen*. The evaluation follows the InterHuman protocol used by the provided codebase and reports both semantic and non-semantic metrics. Semantic quality is measured by FID, diversity, and R-precision when available. Physical plausibility is measured by overlap, penetration volume, foot skating, and maximum acceleration. For multi-person generation, we compare against InterGen with inpainting-based composition and against FreeMotion [1] where available.

Unless otherwise noted, PINO keeps the pretrained generator frozen and applies optimization only at inference time. This matters because improvements cannot be attributed to additional supervised training or changes in backbone capacity.

| Method | Overlap ↓ | PenVol. ($cm^3$) ↓ | Foot Skate ↓ | Max Acc. ↓ |
|---|---|---|---|---|
| GT | 0.029 | 471.75 | 0.099 | 0.074 |
| InterGen | 0.119 | 3112.72 | **0.124** | **0.034** |
| PINO-InterGen | **0.000** | **275.65** | 0.125 | **0.034** |

(a) Non-semantic metrics.

| Method | FID ↓ | Div. → | R-Prec. ↑ (Top3) |
|---|---|---|---|
| GT | 0.983 | 7.921 | 0.715 |
| InterGen | 13.278 | 7.793 | 0.674 |
| PINO-InterGen | **13.163** | **7.904** | **0.675** |

(b) Semantic metrics.

Table 2. Evaluation of two-person interaction generation while avoiding overlap.

| Method | Overlap↓ | PenVol.↓ | Foot Skate↓ | Max Acc.↓ |
|---|---|---|---|---|
| InterGen (inpainting) | 0.766 | 25215.78 | 0.099 | **0.030** |
| PINO-InterGen | **0.000** | **785.01** | **0.091** | 0.031 |

Table 3. Evaluation of multi-person interaction generation

### 4.2. Two-Person Interaction Generation with Overlap Avoidance

We first evaluate whether PINO can improve physical plausibility in the base two-person setting without sacrificing semantic quality. Table 2 shows that it does. PINO-InterGen reduces overlap from 0.119 to 0.000 and penetration volume from 3112.72 to 275.65. At the same time, FID slightly improves from 13.278 to 13.163, diversity increases from 7.793 to 7.904, and R-precision is marginally higher.

These results are important because they isolate the effect of noise optimization from the more complex multi-person setting. The gain is not coming from a different motion prior; it comes from using the same pretrained prior more effectively under a physically informed objective.

### 4.3. Multi-Person Interaction Generation

We next evaluate the main setting of interest: composing larger groups by sequentially adding participants around a pivot. Table 3 compares PINO-InterGen against InterGen with inpainting-based multi-person composition. PINO reduces overlap from 0.766 to 0.000 and penetration volume from 25215.78 to 785.01, while also slightly improving foot skating. Maximum acceleration is nearly unchanged (0.031 vs. 0.030), indicating that the gain does not come from overly damped motions.

To study scaling, Table 4 reports pairwise quality as the group grows from two to five people using person 1 as the pivot. PINO-InterGen consistently achieves the best FID for every pair and dramatically lowers overlap at each step. The advantage becomes larger as more participants are added: for pair $(1, 5)$, FID drops from 19.501 to 16.911 and overlap drops from 0.977 to 0.069 compared with InterGen. Earlier pairs remain near zero overlap, which indicates that sequen-

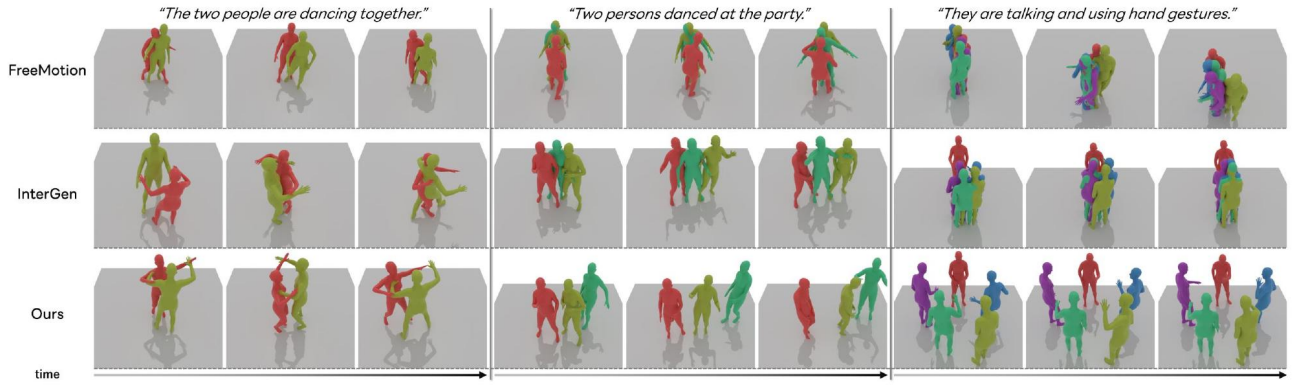


Figure 2. Comparison of the generated multi-person interaction. The visualization highlights that PINO produces cleaner spacing and fewer severe collisions when new participants are added sequentially around a pivot.

| Method | Pair | FID↓ | Overlap↓ | Foot Skate↓ | Max Acc.↓ |
|---|---|---|---|---|---|
| FreeMotion | (1,2) | 19.383 | 0.255 | **0.112** | **0.034** |
| | (1,3) | 25.184 | 0.806 | **0.122** | 0.038 |
| | (1,4) | 25.812 | 0.964 | 0.127 | 0.040 |
| | (1,5) | 25.671 | 0.991 | 0.130 | 0.042 |
| InterGen | (1,2) | 13.100 | 0.119 | 0.126 | **0.034** |
| | (1,3) | 19.414 | 0.799 | 0.125 | **0.036** |
| | (1,4) | 19.386 | 0.937 | **0.124** | **0.038** |
| | (1,5) | 19.501 | 0.977 | **0.124** | **0.039** |
| PINO-InterGen | (1,2) | **12.920** | **0.000** | 0.127 | **0.034** |
| | (1,3) | **17.170** | **0.002** | 0.135 | 0.040 |
| | (1,4) | **16.890** | **0.019** | 0.138 | 0.043 |
| | (1,5) | **16.911** | **0.069** | 0.142 | 0.045 |

Table 4. Evaluation of multi-person interaction generation. From pair (1,2), interactions are generated incrementally using 1 as a pivot. Overlap is among all individuals generated up to that step.

tial composition does not accumulate severe geometric failures as quickly as the baseline.

### 4.4. Long-Duration Motion Extension

We finally evaluate motion extension. In the semantic evaluation of Table 5, PINO-InterGen improves FID from 20.390 to 18.930, reduces overlap from 0.089 to 0.000, lowers foot skating from 0.081 to 0.054, and substantially lowers maximum acceleration from 0.036 to 0.025. In the non-semantic extension benchmark of Table 6, PINO likewise improves every reported metric, including overlap (0.672 to 0.031), penetration volume (15622.16 to 1375.24), and foot skating (0.070 to 0.045).

Together, these results suggest that inference-time optimization is effective not only for adding more people, but also for extending interaction duration while preserving realism at the transition.

| Method | FID↓ | Overlap↓ | Foot Skate↓ | Max Acc.↓ |
|---|---|---|---|---|
| InterGen | 20.390 | 0.089 | 0.081 | 0.036 |
| PINO-InterGen | **18.930** | **0.0** | **0.054** | **0.025** |

Table 5. Evaluation of semantic elements in two-person motion extension. Extended motions are cropped to match the ground truth length for evaluation.

| Method | Overlap↓ | PenVol.↓ | Foot Skate↓ | Max Acc.↓ |
|---|---|---|---|---|
| InterGen (inpainting) | 0.672 | 15622.16 | 0.070 | 0.044 |
| PINO-InterGen | **0.031** | **1375.24** | **0.045** | **0.043** |

Table 6. Evaluation of non-semantic elements in motion extension.

## 5. Analysis

### 5.1. Ablation on Motion Penalties

Table 7 isolates the role of different inference-time penalties. Starting from plain InterGen, adding the root-position loss eliminates positional error but does not address collisions and substantially increases region violations. Adding the overlap loss removes overlap entirely, confirming that the main physical gain in PINO is driven by explicit geometric control rather than being an incidental byproduct of prompt decomposition. Adding the region loss produces the best region-violation score (0.043), while the orientation loss sharply reduces orientation error from 1.0 to 0.208.

This ablation supports the design of Eq. (2): no single penalty solves all control requirements. Instead, the strength of PINO is that these constraints can be combined as needed at inference time, while the pretrained motion prior remains fixed.

### 5.2. Qualitative Observations

Fig. 2 and Fig. 4 provide qualitative evidence consistent with the metrics. Compared with direct multi-person com-

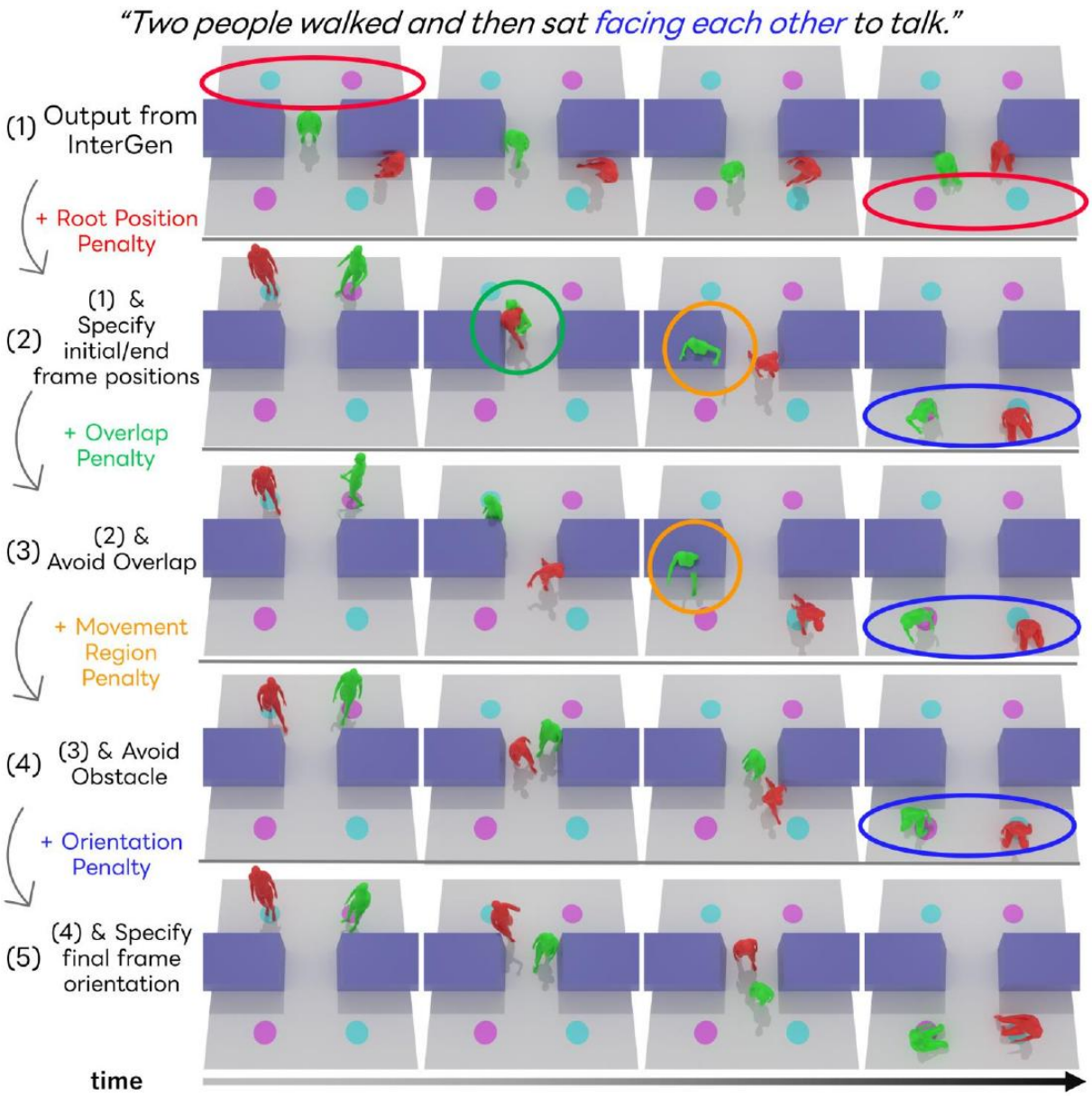


Figure 3. Visualization of the ablation study. Different penalties affect different aspects of the generated motion, with overlap-oriented terms primarily improving physical plausibility and orientation/region terms improving controllability.

position, PINO maintains clearer personal space and produces less entangled body configurations. The visualizations also suggest why pivot-based prompt switching is useful: the new participant can adopt a distinct role relative to the pivot instead of collapsing toward an averaged group behavior.

### 5.3. Discussion

The empirical pattern across all benchmarks is consistent. PINO's largest gains appear in collision-related metrics, especially overlap and penetration volume, and the margin widens as the composition problem becomes harder. Semantic fidelity is preserved in the base two-person setting and improves in motion extension, which indicates that optimizing the initial noise does not simply trade semantics

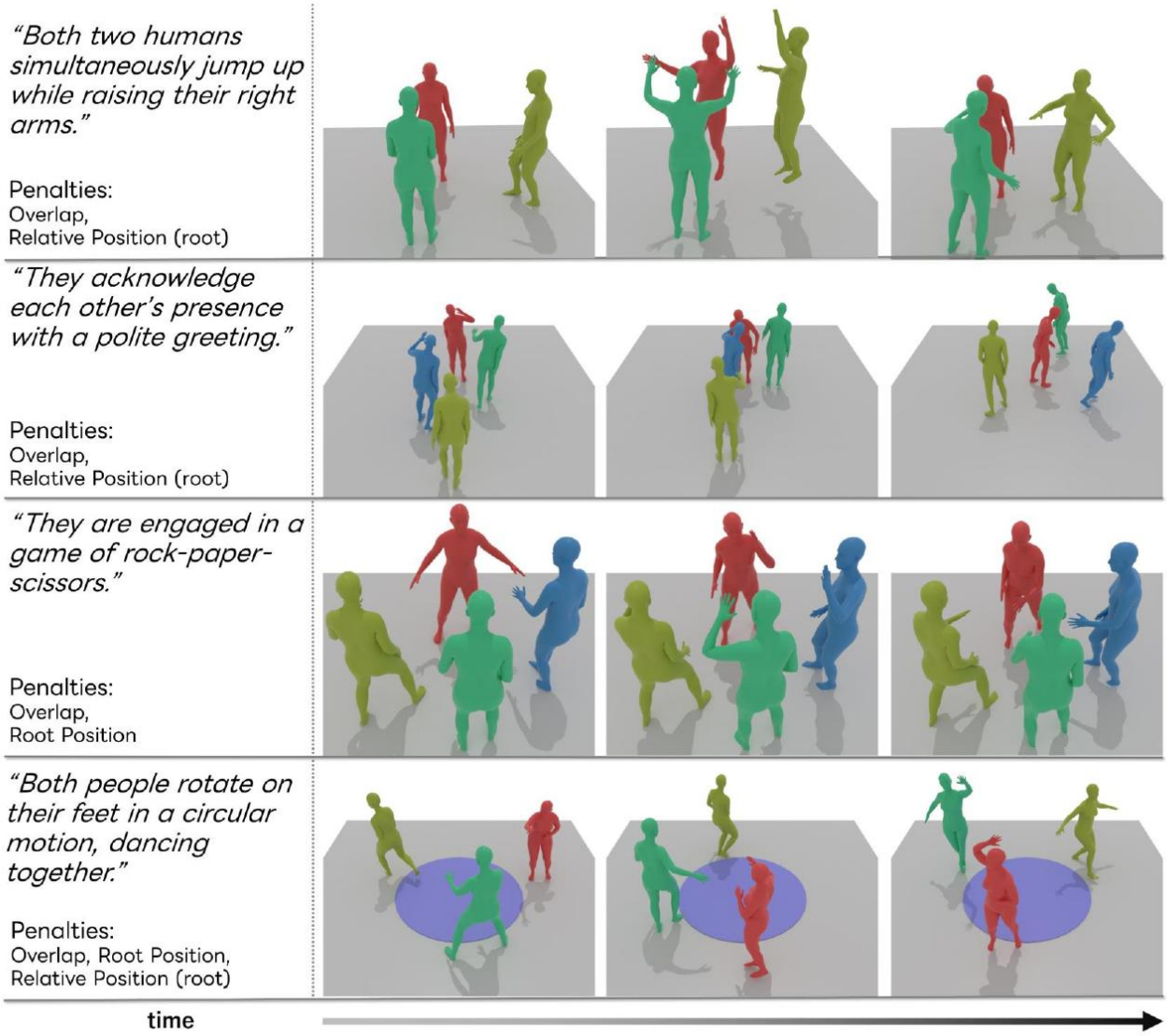


Figure 4. Qualitative results of multi-person interaction generation. PINO supports customizable pairwise interactions within a larger group while maintaining physically cleaner spacing.

for geometry. At the same time, some motion smoothness metrics remain close to the baseline rather than improving dramatically, suggesting that PINO's main benefit is better physical feasibility and controllability rather than a wholesale change in motion style.

## 6. Conclusion

We presented PINO, a training-free framework for arbitrary-sized and long-duration human interaction generation. PINO composes a group around a pivot person using a pretrained two-person diffusion model, and it optimizes the initial diffusion noise with motion-space penalties to enforce physical plausibility and user control. Experiments show that this simple decomposition scales better than direct composition baselines, substantially reduces overlap and penetration, and supports motion extension without retraining. These results indicate that inference-time noise optimization is a strong and practical interface for turning pretrained interaction models into controllable group motion generators.

| Method | Pos. Err.↓ | Overlap↓ | Reg. Viol.↓ | Orient. Err.↓ |
|---|---|---|---|---|
| InterGen | 1.0 | 0.292 | 0.500 | 1.0 |
| + $\mathcal{L}_{\text{root}}$ | **0.0** | 0.333 | 0.917 | 1.0 |
| + $\mathcal{L}_{\text{overlap}}$ | **0.0** | **0.0** | 0.958 | 1.0 |
| + $\mathcal{L}_{\text{region}}$ | 0.083 | **0.0** | **0.043** | 1.0 |
| + $\mathcal{L}_{\text{orientation}}$ | 0.083 | 0.043 | 0.083 | **0.208** |

Table 7. Ablation study on motion penalties. Metrics include positional errors (Pos. Err.↓), overlap (Overlap↓), region violations (Reg. Viol.↓), and orientation errors (Orient. Err.↓). Positional errors are measured with a threshold of 20 cm, region violations with a threshold of 10 cm, and orientation errors with a threshold of 20 degrees. Lower values indicate better performance.

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